L-F-333A
February 26, 1974
SUPERSEDING
Fed. Spec. L-F-333
July 18, 1957

FEDERAL SPECIFICATION

FILM, PHOTOGRAPHIC, MOTION PICTURE (BLACK AND WHITE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the detail requirements for black-and-white motion picture film.

1.2 Classification. Motion picture films covered by this specification shall be alpha-munerically classified, for physical and sensitometric characteristics, in accordance with table I Film Classification System of this specification.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-601	- Box, Wood, Cleated Plywood.
PPP-B-621	- Box, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-640	- Box, Fiberboard, Corrugated, Triple-Wall.

Federal Standard:

Fed. Std. No. 170	- Film Photographic, Black and White, Classification and Testing Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-B-131	- Barrier Material, Water Vaporproof, Flexible.
MIL-L-10547	- Liners, Case, and Sheet, Overwrap, Water-Vaproproof, or Waterproof, Flexible.
MIL-F-26615	- Fixing Bath, Conventional Photographic, Acid Hardening.

Military Standards:

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-147	- Palletized and Containerized Unit Loads 40 inch x 48 inch Pallets, Skids, Runners, or Pallet Type Base.
MS-24320	- Spool-Photographic Film, Standard 35mm.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute (ANSI) Standards:

PH1.25	- Safety Photographic Film, Specifications for (ISO R543).
PH1.29	- Curl of Photographic Film, Methods for Determining.
PH1.31	- Brittleness of Photographic Film, Methods for Determining.
PH1.33	- 16mm 100-foot, 16mm 200-foot, 35mm 100-foot, and 70mm 100-foot, Spools for Recording Instruments and for Microfilm and Still-Picture Cameras, Dimensions for.
PH22.5	- 16mm Motion-Picture Film, 2R-3000, Dimensions for (ISO R69).
PH22.12	- 16mm Motion-Picture Film, 1R3000, Dimensions for (ISO R69).

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American National Standards Institute (ANSI) Standards: (cont'd)

PH22.34	- 35mm Motion-Picture Film, BH1870, Dimensions for
PH22.36	- 35mm Motion-Picture Film, KS-1870, Dimensions for
PH22.37	- Raw Stock Cores for 35mm Motion-Picture Film, Dimensions of (ISO R1039)
PH22.38	- Raw Stock Cores for 16mm Motion-Picture Film, Dimensions of (ISO R1039)
PH22.71	- 32mm Motion-Picture Film, 2R-3000, Dimensions for
PH22.75	- Designation of A and B Windings for Motion-Picture Raw Stock (ISO R1200)
PH22.87	- 100 Mil Magnetic Striping on 16mm Motion Picture Film, Perforated One Edge, Dimensions for (ISO R490, Agrees with ISO R891)
PH22.93	- 35mm Motion-Picture Film, BH-1866, Dimensions for
PH22.101	- Magnetic Striping of 16mm Motion-Picture Film, Perforated 2R-3000, Dimensions for (Agrees with ISO R163)
PH22.109	- 16mm Motion-Picture Film, 1R-2994, Dimensions for
PH22.110	- 16mm Motion-Picture Film, 2R-2994, Dimensions for
PH22.119	- 70mm Motion-Picture Film, Perforated 65mm, KS-1870, Dimensions for
PH22.127	- Magnetic Striping of 16mm Prints, Having Magnetic-Photographic Sound Records, Dimensions of
PH22.137	- Four-Track Magnetic Sound Records for 35mm Release Prints, Dimensions of
PH22.139	- 35mm Motion-Picture Film, KS-1866, Dimensions for
PH22.149	- 8mm Motion-Picture Film, Perforated Super 8, IR-1667, Dimensions for (ISO 1700)

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American National Standards Institute (ANSI) Standards: (cont'd)

PH22.150	- 16mm Motion-Picture Film, Perforated Super 8, 2R-1667 (1-3), Dimensions for
PH22.151	- 16mm Motion-Picture Film, Perforated Super 8, 2R-1664 (1-3), Dimensions for
PH22.159.1	- Super 8 Motion-Picture Film Camera Cartridge and Cartridge-Camera Fit, Specifications for
PH22.161	- Magnetic Striping of Super 8, Motion-Picture Film Perforated 1R-1667, Dimensions of
PH22.162	- Magnetic Striping of 16mm Motion-Picture Film Perforated Super8, 2R-1667 (1-4), Dimensions of
PH22.163	- Magnetic Striping of 16mm Motion-Picture Film Perforated Super 8, 5R-1667, Dimensions of
PH22.169	- 35mm Motion-Picture Film Perforated Super 8 2R-1664 (1-0), Dimensions for
PH22.170	- 35mm Motion-Picture Film Perforated 16mm, 3R-3000 (1-3-0), Dimensions for
PH22.176	- Magnetic Striping of 16mm Motion-Picture Film Perforated Super 8, 2R-1667 (1-3), Dimensions of

(Application for copies should be addressed to American National Standards Institute, 1430 Broadway, New York, NY 10018.)

American Society for Testing and Materials (ASTM) Standards:

D1004-66	- Tear Resistance of Plastic Film and Sheeting, Test for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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TABLE I. FILM CLASSIFICATION SYSTEM

SAMPLE DESIGNATOR = 24 - 33 - 24d

Product	Type	Support	Total Thickness (Mils)	Spectral Sensitivity	Gamma	Relative Sensitivity
	1 - Negative	1 - Cellulose-ester regular	1 - 1.0 - 2.4	1 - Extended ultra-violet	1 - 0.10 / 0.50	a - .57 - 1.1
2 - Motion picture film	2 - Postive film	2 - Cellulose-ester low shrink	2 - 2.5 - 3.8	2 - Blue sensitive	2 - 0.51 / 1.00	b - 1.2 - 2.1
	3 - Reversal	3 - Synthetic Resin	3 - 3.9 - 5.1	3 - Orthochromatic	3 - 1.01 / 1.50	c - 2.2 - 4.2
	4 - Duplicating		4 - 5.2 - 6.5	4 - High Green	4 - 1.51 / 3.00	d - 4.3 - 8.5
	5 - Sound recording		5 - 6.6 - 8.2	5 - Panchromatic	5 - 3.01 / 5.00	e - 8.6 - 17
	6 - Other		6 - 8.3 - 10.0	6 - Panchromatic (extended red)	6 - 5.01 / above	f - 18 - 35
			7 - 10.1 - 12.0	7 - Infrared	7 - Other	g - 36 - 71
			8 - Other			h - 72 - 140
						i - 141 - 280
						j - 281 - 570
						k - 571 - 1100
						l - 1101 - 2300
						m - 2301 - 4500

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(Technical Society and Technical Association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Materials. When materials not specifically designated are used, they shall be entirely suitable for the purpose intended and of high quality,

3.1.1 Radioactive contamination. All materials used in the manufacture, packaging and packing, including the finished product, shall be free from radioactive contamination which would be harmful to the product.

3.2 Film base. The film base shall be an optically homogeneous, transparent medium of uniform thickness and shall be visually neutral except when tinted (see 3.3.1). All films shall be "Safety Photographic Film" as defined in ANSI PH1.25.

3.3 Antihalation protection and base density.

3.3.1 Antihalation protection. Antihalation protection shall be provided in Motion Picture film and shall be in the form of tinted base or coated layer. In either form, the film shall transport through the camera satisfactorily. When a coated layer antihalation protection is used, it shall be bleached or completely removed during processing without interfering with the normal functioning of the processing solutions and without requiring any special manipulation of the film.

3.3.2 Base density.

3.3.2.1 Tinted base. Tinted base density shall be within 0.07 to 0.12 for low-tint base and within 0.20 to 0.30 for high-tint base.

3.3.2.2 Clear base. When tinting is not used, the base density shall not exceedd 0.06.

3.3.2.3 Reversal films. Unless otherwise specified reversal films (Type 3) shall be supplied on a base having a density of 0.10 to 0.30.

3.3.2.3.1 Maximum density. Maximum density of reversal films shall be at least 2.1 when processed in accordance with Fed Std-170.

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3.4 Physical defects. The film shall be free from scratches and pin holes to the degree that they will not interfere with the intended use. After being tested as specified in 4.4.2, film shall not show streads, spots, aor areas of marked higher or lower sensitivity than the surrounding areas.

3.5 Film width and perforationgs. Dimensions and tolerances of the films shall conform to the following ANSI Standards as specified by the procuring activity (see 6.2):

Super 8mm	
PH22.149	- 8mm Motion-Picture Film, Perforated Super 8, 1R-1667, Dimensions for
16mm	
PH22.5	- 16mm Motion-Picture Film, 2R-3000, Dimensions for (ISO R69)
PH22.12	- 16mm Motion-Picture Film, 1R-3000, Dimensions for (ISO R69)
PH22.109	- 16mm Motion-Picture Film, 1R-2994, Dimensions for
PH22.110	- 16mm Motion-Picture Film. 2R-2994, Dimension for
PH22.150	- 16mm Motion-Picture Film, Perforted Super 8, 2R-1667 (1-3), Dimensions for
PH22.151	- 16mm Motion-Picture Film, Perforated Super 8, 2R-1664 (1-3), Dimensions for
32mm	
PH22.71	- 32mm Motion-Picture Film, 2R-3000, Dimensions for
35mm	
PH22.34	- 35mm Motion-Picture Film, BH-1870, Dimensions for
PH22.36	- 35mm Motion-Picture Film, KS-1870, Dimensions for
PH22.93	- 35mm Motion-Picture Film, BH-1866, Dimensions for
PH22.139	- 35mm Motion-Picture Film, KS-1866, Dimensions for
PH22.169	- 35mm Motion-Picture Film, Perforated Super 8, 2R-1664 (1-0), Dimensions for
PH22.170	- 35mm Motion-Picture Film, Perforated 16mm, 3R-3000 (1-3-0), Dimensions for
70mm	
PH22.119	- 70mm Motion-Picture Film, Perforated 65mm, KS-1870, Dimensions for

3.6 Film length. The film shall be supplied in lengths as specified by the procuring activity (see 6.2c).

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3.7 Film winding. Super 8mm film shall be wound emulsion side outward on cartridge and 16mm and 35mm films shall be wound emulsion side inward on spools or cores as specified by the procuring activity (see 6.2f). Table II shows the military designaion for the different spools, cores and cartridges along with equivalent commerical designations, where known.

TABLE II Spool and Core Designations

Spool Number	Film Width (mm)	Length (ft)	Standard	Commercial Designation GAF	Kodak	Dupont
1	(Deleted)					
2	35mm	100	MS-24320-2	649620	No. 10	No. 16
3	16mm	50		649205	EK 16mm	
4	16mm	100	ANSI PH1.33	649216	16mm R-90	44
5	16mm	200	ANSI PH1.33	649225	EKR-94	49
6	16mm	400		649230		No. 60
Core Number						
1	35mm		ANSI PH22.37	335230	EK-U	24
2	35mm			Universal	Universal	Universal
3	(Deleted)					
4	35mm			335220	EK-AA	4
5	16mm		ANSI PH22.38	335120	EK-T	32
6	16mm			335114		
7	16mm		ANSI PH22.38	335130	EK-Z	
Cartridge Number						
1	Super 8mm	50	ANSI PH22.159.1			

3.7.1 Sound recording film. Unless otherwise specified (see 6.2e), super 8mm, 16mm, and 35mm sound recording film, perforated on one edge only in accordance with ANSI PH22.109, shall be supplied with winding commercially known as "Winding A" or "Winding B" in accordance with ANSI PH22.75. When the film is wound on a flanged spool containing one round and one square spindle hole, the hole in the flange on the side of the perforation shall be as follows:

Winding A: Round hole
Winding B: Square hole

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3.8 Physical characteristics. Unless otherwise specified, the film shall have the following characteristics to insure suitability for use in motion picture cameras.

3.8.1 Thickness. The total thickness of the sensitized film for the specified designator shall fall within the limits indicated in table I, when the film is measured at equilibrium with a temperature of 70° F and a relative humidity of 50 percent (see 4.4.3.1).

3.8.2 Tensile strength. The film shall have a minimum tensile strength, in the lengthwise direction, as follows (see 4.4.3.2):

Support type (See Table I)	Tensile Strength (PSI Minimum)
1	10,000
2	9,000
3	20,000

3.8.3 Tearing resistance. When tested as specified in 4.4.3.3 the film shall have a minimum tearing resistance in the lengthwise direction, as follows:

Support Type (See Table I)	Tearing Resistance (Kilograms Minimum)
1	2.0
2	2.0
3	4.5

3.8.4 Rapid processing. Unless otherwise specified the film shall be without physical degradation when processed and tested in accordance with 4.4.3.4.

3.8.5 Curl. When tested as specified in Method C of ANSI PH1.29, the maximum curl of the film shall not exceed a munerical value of 100 before processing and after processing.

3.9 Sensitometric characteristics. The sensitometric characteristics of the developed films shall conform to Table III.

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Table III. Sensitometric Characteristics

Designator (Note 1)	Exposure		Development		Relative Sensitivity		Avg. Gradient (see 6.3.3)	Gamma	A D (see 6.3.4)	Filter Factors				Fog (Max)	Incub. Fog (Max)
	Illuminant #	Time	Formula (Note 2)	Time	Method	Limits				25	47	58	898		
21-XX-34j	2	1/20-1/80	S5.3.1.1	8	A	300-600		1.8-2.5	.75-.85		7	7		0.10	0.15
21-XX-34l	2		S5.3.1.1	8	A	1000 Min		1.8-2.5	.75-.85		10	6		0.10	0.20
21-XX-52c					B										
21-XX-52f	1	1/20-1/80	S5.3.1.3	4-8	A	20-50			.75-.85	12	8	12		0.05	0.10
21-XX-52g	1	1/20-1/80	S5.3.1.3	3-8	A	30-65			.75-.85	9	9	7		0.07	0.10
21-XX-52g	1		S5.3.1.5	5-8	A	60-100		7.5-9.5	.75-.85	12	8	14		0.07	0.20
21-XX-52h	1	1/20-1/80	S5.3.1.2	5-10	A	50-100			.75-.85	12	8	12		0.05	0.10
21-XX-52h	1		S5.3.1.2	3-10	A	50-100			.75-.85	16	8	12		0.05	0.10
21-XX-52i	1		S5.3.1.2	5-10	A	80-225		.70	.75-.85	12	8	12		0.05	0.10
21-XX-52i	1		S5.3.1.2	3-10	A	80-225			.75-.85	12	8	12		0.05	0.20
21-XX-52j	1		S5.3.1.2	5-12	A	250-500			.75-.85	14	8	14		0.10	0.20
21-XX-52j	1	1/20-1/80	S5.3.1.7	5-12	B	250-500				12	8	12		0.10	0.20
21-XX-52k	1	1/20-1/80	S5.3.1.1	6-12	A	400-800		1.8-2.5	.75-.85	8	8	16		0.10	0.20
21-XX54i	1		S5.3.1.1	6-12	A	125-250		1.8-2.8	.75-.85	8	8	16		0.10	0.20
21-XX-64i	1	1/20-1/80	S5.3.1.1	8	A	125-250		1.8-2.5	.75-.85	6	12	8		0.07	0.10
21-XX-64i	1		S5.3.1.1	8	A	125-250		1.8-2.5	.75-.85	6	12	15		0.10	0.10

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Table III. Sensitometric Characteristics (cont)

Designator (Note 1)	Exposure		Development		Relative Sensitivity		Avg. Gradient (see 6.3.3)	Gamma	A D (see 6.3.4)	Filte Factors				Fog (Max)	Incub. Fog (Max)
	Illum-inant #	Time	Formula	Time	Method	Limits				25	47	58	89B		
21-XX-72c	3	1/20-1/80	S5.3.1.1	4-8	A	2-5			.75-.85				3.0	0.10	0.15
21-XX-72c	3		S5.3.1.3	4-7	A	9-18		0.70	.75-.85				3.0	0.07	0.14
21-XX-72g	3		S5.3.1.2	3-6	A	30-60		0.70	.75-.85				2.4	0.10	0.20
21-XX-74g	3	1/40-1/120	S5.3.1.1	8	A	30-95		1.8-2.7	.75-.85				2.0	0.15	0.35
21-XX-62k	1	1/20-1/80	S5.3.1.1	6-12	A	40C-800		0.70	.75-.85	6	9	15		0.10	0.15
22-XX-23b	2	1/40-1/120	S5.3.1.5	6-12	B	1-2		1.4						0.06	0.07
22-XX-24a	2		S5.3.1.5	3	B	0.8-1.6		1.8-2.3						0.02	0.04
22-XX-24b	2	1/40-1/120	S5.3.1.5	3-6	B	1-3		2.5						0.03	0.04
22-XX-24c	2		S5.3.1.5	3	B	3-6		2.4-3.0						0.02	0.04
22-XX-24d	2		S5.3.1.5	3	B	6-9		1.7-2.2						0.02	0.04
22-XX-34l	2		S5.3.1.5	8	B	1000 Min		1.8-2.5			10	6		0.10	0.20
22-XX-54c	2	1/40-1/120	S5.3.1.5	6-12	A	1.8-3.5		0.70	.75-.85					0.05	0.06
23-XX-52g	1		S5.3.1.7	3-8	B	32-80				12	8	12		0.05	0.10
23-XX-52i	1		S5.3.1.2	4-10	B	100-250				12	9	12		0.10	0.20
23-XX-52i	1		S5.3.1.2	4-10	B	100-250		1.25-1.6		12	9	14		0.10	0.20

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Table III. Sensitometric Characteristics (cont)

Designator (Note 1)	Exposure		Development		Relative Sensitivity		Avg. Gradient (see 6.3.3)	Gamma	A D (see 6.3.4)	Filte Factors				Fog (Max)	Incub. Fog (Max)
	Illum-inant #	Time	Formula	Time	Method	Limits				25	47	58	89B		
23-XX-52k	1		S5.3.1.2	6-10	B	60-1000				12	9	12		0.10	0.02
23-XX-53i	1	1/20-1/80	S5.3.1.7		B	110-225	.95-1.25			9	9	7			
23-XX-53k	1	1/20-1/80	S5.3.1.7		B	400-800	1.1-1.4			9	9	7			
24-XX-24c	2		S5.3.1.1	3	B		1.7-2.2							0.02	0.04
24-XX-52c	2		S5.3.1.5	4-9	B	2-4				8	12	15		0.02	0.04

Note 1. The Table uses XX to represent Support Type and Total Thickness as these Characteristics have no bearing on purely photographic values.

Note2. See Fed Std No. 170 for applicable formula cited by paragraph number.

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3.10 Magnetic stripe. Film shall be supplied with a magnetic stripe in accordance with the following ANSI Standards when specified by the procurement activity:

8mm film	
PH22.161	Magnetic Striping of Super 8 Motion-Picture Film Perforated 1R-1667, Dimensions of
16mm film	
PH22.87	100-MIL Magnetic Striping on 16mm Motion Picture Film, Perforated One Edge, Dimensions for (ISO R490 Agrees with R891)
PH22.101	Magnetic Striping of 16mm Motion-Picture Film, Perforated 2R-3000, Dimensions of (Agrees with ISO R163)
PH22.127	Magnetic Striping of 16mm Prints, Having Magnetic-Photographic Sound Records, Dimensions of
PH22.162	Magnetic Striping of 16mm Motion-Picture Film Perforated Super 8, 2R-1667 (1-4), Dimensions of
PH22.176	Magnetic Striping of 16mm Motion-Picture Film Perforated Super 8, 2R-1667 (1-3), Dimension of
35mm film	
PH22.137	Four-Track Magnetic Sound Records for 35mm Release Prints, Dimensions of
PH22.163	Magnetic Striping of 35mm Motion-Picture Film Perforated Super 8, 5R-1667, Dimensions of

3.11 Edge marking. Unless otherwise specified by the procuring activity (see 6.2i), all negative motion-picture film shall be edge marked. Commercial nomenclature and sequential edge numbers shall be marked within 0.098 inch of the edge of the film. The edge numbers shall appear at regular intervals of not more than one foot and in ascending order toward the core, except when the film is supplied single perforated, winding A, in accordance with ANSI PH22.75.

3.12 Brittleness. When tested in accordance with ANSI PH1.31, the value obtained shall not exceed 0.275 inch, with readings based on the first sign of a crack or tear.

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3.13 Expiration date. The material supplied in accordance with this specification shall be marked with expiration dates of 27 months from the time of delivery except for those designators listed below which shall be marked as specified.

21-14-72e	= 18 months	23-15-52g	= 30 Months
21-14072g	= " "	23-33-52g	= " "
		23-14-52j	= " "
		23-14-52i	= " "

3.14 Workmanship. Workmanship shall be in accordance with high grade commercial practice covering the manufacture of sensitized photographic materials of the gelatin-silver halide type.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection before preparation for delivery. The contractor shall perform the inspections specified in 4.2.2 through 4.3. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.2.1 Alternative inspection procedures. Alternative inspection procedures and inspection equipment may be used by the contractor provided such procedures and equipment will assure quality equal to or better than that obtained by the inspection procedures specified herein and provided the alternative procedures are approved by the procuring activity.

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4.2.2 Inspection of materials and components. The contractor shall assure that materials and components are manufactured, inspected and tested in accordance with the requirements of subsidiary specifications and standards referenced herein. In the event of conflict, this specification shall govern.

4.2.3 Sampling for inspection. Sampling for inspection shall be in accordance with MIL-STD-105. Each lot shall be sampled. A lot shall consist of all materials of the same designator, and each batch and kettle number, included therein, submitted for delivery at one time. The lot size shall be expressed in units of rolls for all inspections, except 4.7 which shall be expressed in units of shipping containers. No more than three (3) rolls shall be drawn from any one shipping container for each lot.

4.2.4 Inspection of the end item. The end item shall be inspected for examination listed, using inspection levels and acceptable quality levels (AQL's) set forth.

Inspection	Inspection Level	AQL
4.2.4.1	S-3*	2.5dphu
4.2.4.2		2.5dhpu
4.2.4.3	S-2	2.5dphu

*The same sample rolls are to be used for 4.2.4.1 and 4.2.4.2.

4.2.4.1 Examiniation of the end item for defects in appearance and workmanship. The sample unit for this examination shall be nine linear inches of sensitized film from each sample roll.

Examine	Defect
Appearance and Workmanship	Film not completely coated. Film unevenly coated. Emulsion surface shows scuffs, scratches, foreign matter, or abrasions. Holes, tears, cuts, cracks, or sharp creases. Perforation holes not cleanly punched.

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4.2.4.2 Examination of end item for dimensional defects. The sample unit for this examination shall be two and one-half linear feet of film from each sample roll.

Examine	Defect
Width	Not specified width.
Perforations	Film perforations do not conform to applicable ANSI standard.

4.2.4.3 Examination of the end item for defects in construction and length and blocking. The sample unit shall be one roll of film. This examination shall consist of the following:

Examine	Defect
Film	Roll not evenly or neatly wound.
	Not wound on specified core or spool.
	Core or spool sizes not as specified.
	Core or spool cracked, broken, chipped, mutilated or collapsed.
	Not spooled as specified with emulsion in proper position.
	Roll not continuous.
	Roll contains more than one splice.
	Splices poorly made.
	Film blocks.
Length of roll	Not in conformance with length specified in contract.

4.3 Testing of the end item. Testing of the end item shall be conducted in accordance with Tabel IV for the characteristics as indicated therein on each lot submitted for acceptance. Three sample units each drawn from a different shipping container, shall be tested with no evidence of failure to meet test requirements.

4.3.1 Test procedures, equipment and terms. Except as specified herein, the test procedures, equipment and terms shall be in accordance with Fed. Std. No. 170.

4.4 Test methods.

4.4.1 Base density. The base density of the film shall be determined as specified in Fed. Std. No. 170.

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4.4.2 Physical defects. Two pieces of film each equal to nine square inches shall be exposed in accordance with physical defects test of Fed. Std. No. 170 and visually examined to determine compliance with the requirement specified in 3.4.

4.4.3 Physical characteristics. Unless otherwise specified by the procuring activity, the following physical tests shall be conducted. The test samples shall be in equilibrium with temperatures of 70° F and relative humidity of 50 percent.

4.4.3.1 Film thickness. The thickness of the film shall be measured by a micrometer or any instrument capable of measuring thickness within 0.0001 inch. Tests shall be made on the master roll at a minimum of four points; one at the beginning, one at the end and one on each side of the roll.

4.4.3.2 Tensile strength test. The test procedure used for measuring the tensile strength of the film shall be in accordance with Fed. Std. No. 170. Ten samples cut from the lengthwise direction of the film shall be tested.

4.4.3.3 Tearing resistance test. The test procedure for measuring the tearing strength of the film shall be in acordance with ASTM D1004-66. Test samples shall be cut from raw film stock before final slitting and perforating. Ten samples cut from the lengthwise direction of the film shall be tested.

4.4.3.4 Rapid processing physical endurance test. This test shall consist of processing the test samples through a series of 3 tanks containing the solutions listed below. The sequence, temperature and time shall be as indicated. Each tank shall be constructed to provide agitation within the tank. Softening, peeling, reticulaion or partial loss of emulsion shall be cause for rejection.

a. Development for 30 seconds at 110° F in Armed Forces Developer No. 1 (formulated in accrodance with paragraph S5.3.1.1, Fed. Std. No. 170).

b. Short stop for 1 minute at solution temperature of 100° F in 2 percent solution of acetic acid in distilled water.

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Table IV. Instructions for testing (sample unit)

Characteristic	Specification Reference		Rqmts Appl To	Number Determinations For Sample Unit	Results Rept As	
	Requirement	Test Method	Indv Unit		Pass or Fail	Numerically To Nearest
Base Density 1/	3.3.2	4.4.1	X	Average of 2	X	.01 Density
Physical Defetcs 1/	3.4	4.4.2	X	1	X	
Physical Characteristics (when applicable) 1/	3.8	4.4.3				
Film Thickness 1/	3.8.1	4.4.3.1	X	1	X	
Tensile Strength 1/	3.8.2	4.4.3.2	X	1	X	
Tearing Resistance 1/	3.8.3	4.4.3.3	X	1	X	
Rapid Processing 1/	3.8.4	4.4.3.4	X	1	X	
Curl 1/	3.8.5	4.4.4	X	1	X	
Brittleness 1/	3.12	4.4.5				
Sensitometric Chracter-istics	3.9	4.5				
Relative Sensitivity	3.9	4.5	X	Average of 2		Speed Gp
Gamma	3.9	4.5	X	Average of 2		.01 Gradient
Filter Factor 2/	3.9	4.5	X	Average of 2		Speed ratio to the nearest integer
Max Fog	3.9	4.5	X	Average of 2		.01 Density
Max Incubation Fog	3.9	4.5	X	Average of 2		.01 Density

Note: 1/ Contractor may furnish certificate of compliance stating that, in previous tests conducted on the same product and type of material, the material conforms to these requirements.

Note: 2/ Filter factors should be based on actual filter difference and shall not be rounded off to the nearest ASA speed group.

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c. Fix for 30 seconds at 100° F in a fixer solution that is based on and performs in accordance with MIL-F-26615.

d. A minimum of 10 minute wash in running water at a temperature of 100° F.

4.4.4 Curl. Using test specimens cut from unprocessed and processed samples, conduct the curl test in accordance with Method C of ANSI PH1.29.

4.4.5 Brittleness. Using specimens cut from samples, conduct the wedge brittleness test in accordance with Method B of ANSI PH1.31.

4.5 Sensitometric characteristics. The sensitometric tests shall be conducted in accordance with the applicable test methods of Fed. Std. No. 170 and the test conditions specified in Tabel III. The developer solution shall be used only during a period of from 2 to 12 hours after mixing.

4.5.1 Magnetic stripe. The magnetic stripe shall be measured to determine compliance with 3.10.

4.6 Water vapor resistance test. Four packages of the same designator and emulsion number shall be taken. The relative humidity with which the film in two of these packages is in equilibrium shall be determined using the method described in 4.6.1. The remaining two packages shall then be placed in an atmosphere having a temperature of 90° F ± 2° F and a relative humidity of 90 to 95 percent for a period of 3 months, after which they shall be removed, opened, and the relative humidity with which the film is in equilibrium determined in accordance with the procedure set forth in 4.6.1. The average relative humidity with which the film is in equilibrium shall not be increased by more than 10 percent relative humidity when calculated by taking the difference between the average equilibrium relative humidity obtained for the two sets of samples. For example, if the packaged film were originally in equilibrium with an atmosphere of 50 percent relative humidity, it shall be in equilibrium with an atmosphere having a relative humidity, not above 60 percent after the storage treatment.

4.6.1 Determination of the equilibrium relative humidity of packaged film. The package in question shall be opened in a sampling room maintained at a relative humidity of 50 ± 5 percent and the outer layers of film discarded. Three samples of film weighing approximately 5 grams each shall then be selected from representative positions, placed immediately in ground-glass stoppered weighing bottles and weighed. The film shall be

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handled with gloves and shall not be exposed to the atmosphere for more than 10 seconds. Care shall also be taken to avoid breathing on the film. The samples of film shall then be conditioned for 4 hours in an air stream having a velocity of at least 1/10 foot per second and weighed at each of two different known relative humidities at a temperature of 70° F. The two conditioning relative humidities shall not be more than 15 percent relative humidity apart and shall be selected to fall, one above and one below, the equilibrium relative humidity of the packaged film. The average weight of the conditioned film samples shall be plotted against the condition relative humidity, and the two points connected by a straight line, which thus approximates the moisture equilibrium curve of the film over the range in question. The equilibrium relative humidity of the packaged film shall be determined by interpolating the original average weight of the film samples.

4.6.2 The test specified in 4.6 shall be performed on the first lot of containers from each manufacturer. In lieu of subsequent testing, provided there is no change in design or material, a certificate of conformance, certifying that the containers offered have met the test requirements, shall be provided to the authorized Government representative. Test data on which the certificate is based will be made available to the Government representative upon demand.

4.7 Quality conformance inspection of preparation for delivery. An examination shall be made to determine that marking, materials, workmanship, contents and weights comply with the requirements of Section 5 of this specification. Defects shall be scored in accordance with the classification below. The examination shall conform to the inspection procedures of MIL-STD-105 using the special inspection levels. The AQL shall be 4.0 DPHU and the inspection level shall be S-2. The sample unit shall be one shipping container, fully packed, selected just prior to closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects.

Examine	Defect
Packaging and preservation	Not as specified; not in accordance with contract requirements. Unit packages not uniform. Containers of improper construction.
Packing	Not packed as specified. Packing material not as specified; closures not accomplished by specified or required methods or materials. Number of rolls per container, less than indicated.

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Examine	Defect
Markings (Interior and Exterior)	Interior labels not correct color. Data on labels not in accordance with contract specification. Labels not properly attached. Exterior labels or markings illegible, incorrect, incomplete, or not in accordance with contract requirements.
Weight (Interior and Exterior)	Gross or net weight exceeds contract requirements.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be Level A or C as specified (see 6.2j).

5.1.1 Level A.

5.1.1.1 Unit packaging. Film shall be packaged in a light-tight, water-proof container sealed with a water-vapor proof tape. The sealed container shall meet the requirements of the water-vapor resistance test in 4.6, 4.6.1 and 4.6.2. Unit quantities shall be one roll each. All materials used in preservation and packaging of film shall be photographically inert to a degree in proportion to their proximity to the film.

5.1.1.2 Magazines. Each magazine shall be sealed in a moisture impervious envelope conforming to MIL-B-131 and packed in a box conforming to PPP-B-636, Class domestic, style optional. Closure of boxes shall be in accordance with the appendix to PPP-B-636.

5.1.1.3 Intermediate packaging. When intermediate packages are required, place a quantity of packaged film, as specified in 5.1.1.1 within a close-fitting fiberboard box conforming to PPP-B-636, W6c. Closure shall be as specified in the appendix of the box specification. All unit packages within the intermediate package shall bear the same stock number.

5.1.2 Level C. Film shall be packaged in a manner that will afford adequate protection against deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2j).

5.2.1 Level A. Film packaged, as specified in 5.1, shall be packed within close-fitting fiberboard boxes conforming to PPP-B-636, class weather resistant, or in overseas type wooden boxes conforming to PPP-B-601 or PPP-B-621. Exterior containers shall be of uniform shape and size, be of minimum cube and tare, be consistent with the protection required, and containe identical quantities. For PPP-B-636, the gross weight of each pack

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shall be limited to 90 pounds. For PPP-B-601 and PPP-B-621, the gross weight of each pack shall be limited to 200 pounds. Container strength and design shall be adequate to insure protection against damage during shipment, handling, indeterminate storage, and world-wide, redistribution. Closure shall be in accordance with the applicable container specification or appendix thereto, except boxes conforming to PPP-B-636 shall be closed and waterproof sealed in accordance with the appendix to the box specification. Containers in accordance with PPP-B-636 shall meet the waterproofing requirements of paragraph 30.4 and, in this case only, at the option of the manufacturer, the intermediate package specified in 5.1.1.3 may be omitted. Containers in accordance with PPP-B-601 or PPP-B-621 shall be provided with a waterproof case liner complying with MIL-L-10547. These waterproof case liners may bo omitted if unit or intermediate containers comply with PPP-B-636, class weather resistant. As an alternate and at the option of the manufacturer, films packed in accordance with paragraph 5.2.2 may be overpacked to meet the requirements of paragraph 5.2.1.

5.2.2 Level B. Film packaged in accordance with paragraph 5.1 shall be packed in domestic type exterior containers meeting PPP-B-636, class domestic; or PPP-B-640, class 1, style E. Exterior containers shall be of minimum cube and tare, consistent with the protection required. Containers will be of uniform shape and size and contain identical quantities. For PPP-B-636, the gross weight of each pack shall be limited to 90 pounds, and for PPP-B-640, limited to 200 pounds. Closure shall be in accordance with the applicable container specification or appendix thereto. Container strength and design shall be adequate to insure protection against damage during multiple domestic shipment, handling, and covered storage.

5.2.3 Level C. Film packaged, as specified in 5.1, requiring overpacking for acceptance by the carrier, shall be packed in exterior type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall meet the National Motor Freight Classification rules or the Uniform Freight Classification rules.

5.2.4 Palletization. When specified by the procuring activity, exterior containers will be palletized in accordance with MIL-STD-147, or as otherwise specified by the procuring activity. The method of bonding the palletized load will be limited to strapping.

5.2.5 Refrigerated transportation. Unless otherwise specified, shipments will be made under refrigeration (maximum temperature of 50° F.) or other conditions that will insure maintenance of product quality while in transit to destination.

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5.3 Labeling. All films covered by this specification, except 16mm motion picture film short rolls (200 feet or less) and films supplied in magazines, shall have a label on both the unit and intermediate container containing essentially the same information as shown in Figure 1. Rectangular labels may be used in place of the disk label shown. Magazines shall bear a rectangular label on the unit and intermediate package which provides the same information as shown in Figure 1. All labels shall be affixed to the container with a suitable adhesive which will hold the label firmly and which will resist the loosening effect of dampness. The label shall also show the number or rolls inside the metal container, if more than one roll is placed within a single unit container.

5.3.1 Color of labels. Labels shall be printed in heavy type with black ink upon colored paper stock as shown below for the spectral sensitivity designators.

Spectral Sensitivity	Color of Labels
2 - Blue sensitive	White
3 - Orthochromatic	Red
5 - Panchromatic	Green
7 - Infrared	Light Blue

Colors shall permit easy reading of the print.

5.3.2 Label data. The manufacturer's emulsion number shall be stamped on the labels of both the unit and intermediate package where both are used. The applicable designator, stock, manufacturer's name and commercial designation and the military expiration date shall appear on the label. The military expiration date shall be, as specified in 3.13.

5.4 Marking. Each exterior container shall be marked in accordance with MIL-STD-129 or Fed. Std. No. 123, as specified, except that the dimensions of the container and preservation method need not be included. Abbreviated nomenclature may be used when necessary. Unless otherwise specified by the procuring activity, the complete film classification number, emulsion numbers, size and military expiration date shall be marked on the shipping containers. The nomenclature shall be: Film, Photographic, Motion-Picture, Black and White.

5.4.1 Other markings. The following precautionary marking shall be permanently, prominently and legibly marked in a vivid color, preferably red, and in bold faced letters:

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MANUFACTURER'S NAME

TRADE NAME OF FILM CONTAINED

FEET	METERS		EMULSION NUMBER
100	30.5	35MM	

21-14-52i

PERFORATION-ANSI-BH-1870

MILITARY EXPIRATION DATE 200 EXPOSURE INDEX

NUMBER OF ROLLS (If more than one contained)

FILM, PHOTOGRAPHIC, MOTION PICTURE, BLACK AND WHITE

SPECIFICATION: L-F-333A

STOCK NUMBER

This space to contain other pertinent data such as winding A or B in Single Perforated 16MM Film.

Manufacturer's Disclaimed Clauses, Etc.

FIGURE 1

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PHOTO SENSITIVE MATERIALS
STORE IN COOL (MAX. 50° F) DRY (50% MAX.
HUMIDITY) LOCATION AWAY FROM RADIOACTIVE
MATERIALS

6. NOTES

6.1 Intended use. The films covered by this specification are intended for use in motion picture cameras, sound recorders, printers and projectors. The motion picture camera films are intended for use in cameras operating no faster than 128 frames per second, except when specified with fast cycling perforations.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Film classification system designator. (See 1.2)
(c) The film width and type perforations required. (See 3.5)
(d) The film length and quantity required. (See 3.6)
(e) Whether "Winding A" or "Winding B" is required for film (See 3.7.1)
(f) The type of spool, core or cartridge required. (See 3.7)
(g) If rapid processing will be required. (See 3.8.4)
(h) If magnetic striping of the film is required. (See 3.10)
(i) Edge marking requirements. (See 3.11)
(j) Level of packaging, packing, and marking (Level A, level B, or level C). (See Section 5)
(k) Refrigerated transportation. (See 5.2.4)

6.3 Definitions.

6.3.1 Film classification system designator. The film classification system designator is an alpha-numerical system used to classify a given film in accordance with Table I and as shown in the example below:

Product	Type	Support Type	Total Thickness (Mils)	Spectral Sensitivity	Gamma	Relative Sensitivity
2	1	1	4	5	2	1

Reading left to right 21-14-52i

2 - Motion Picture Film
1 - Negative
1 - Cellulose-Ester Regular Base
4 - Thickness Range 5.2 to 6.5 mils
5 - Panchromatic
2 - Gamma range 9.51 to 1.00
i - Relative sensitivity 141 to 280

(When ordering, the full film classification designator must be used.)

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6.3.2 Synthetic resin. Synthetic resin support types include bases such as polyester.

6.3.3 Average gradient. The average gradient is the slope of a straight line between points E_1 and E_2 on the H and D curve. Point E_1 is at a total density of 0.5 and E_2 is at a total density of 2.0.

6.3.4 ΔD. D is the density difference between points E_1 and E_2 on the H and D curve. Point E_1 is a point on the curve where the density is 0.10 above base plus fog density. Point E_2 is a point on the curve where the log exposure is 1.30 greater than the log exposure at point E_1.

6.4 Supersession data. This specification includes the requirements of MIL-F-26386A, 29 April 1960, covering Film, Photographic, Motion Picture, Aerial, Black-and-White.

6.5 Cross reference of classification. The film classification has been completely revised. The items covered by this specification may be binned with prior items. The film classification system specified in this specification will replace the previous system of designating classification under L-F-333 and MIL-F-26386A. The main differences between the previous specifications are shown in the cross reference below.

Previous Specification		This Specification
Type	is now	Product
Sub-type	is now	Support type
Classes	is now	Spectral sensitivity

In addition, other classifications have been added (See Table I of this specification) such as Type, Total Thickness, Gamma, and Relative Sensitivity which will allow users to select a complete designator for films of their choice to meet their particular requirements.

6.6 International standardization agreement. Certain provisions of this specification are the subject of international standardization agreements STANAG Number 3179 and ASCCAS 52/6A. When amendment, revision, or cancellation of this specification is proposed which will effect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardizations offices, if required.

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MILITARY INTERESTS:

Custodians:

Army - EL
Navy - AS
Air Force - 70

Review Activities

Army - MI
Navy - AS
Air Force - 70, 85
Other - NSA

User Activities:

Navy - EC, YD, MC

Preparing Activity:

Army - EL

Coordination method Code-C
Project Number 6750-0268

CIVIL AGENCY COORDINATING ACTIVITIES:

VA - DMS
POSTAL - POS
USDA - AFS
COMMERCE - NBS
HEW - HSM
GSA - FSS

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